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Keeping walls straight: data model and training set size matter for deep learning in building generalization

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ABSTRACT

Deep learning-backed models have shown their potential of conducting map generalization tasks. However, pioneering studies for raster-based building generalization encountered a common “wobbly-wall effect” that makes the predicted building shapes unrealistic. This effect was identified as a critical challenge in the existing studies. This work proposes a layered data representation model that separately stores a building for generalization and its context buildings in different channels. Incorporating adjustments to training sample generation and prediction tasks, we show how even without using more complex deep learning architectures, the widely used Residual U-Net can already produce straight walls for the generalized buildings and maintains rectangularity and parallelism of the buildings very well for building simplification and aggregation in the scale transition from 1:5,000 to 1:10,000 and 1:5,000 to 1:15,000, respectively. Experiments with visual evaluation and quantitative indicators such as Intersection over Union (IoU), fractality, and roughness index show that using a larger input tensor size is an easy but effective solution to improve prediction. Balancing samples with data augmentation and introducing an attention mechanism to increase network learning capacity can help in certain experiment settings but have obvious tradeoffs. In addition, we find that the defects observed in previous studies may be due to a lack of enough training samples. We thus conclude that the wobbly-wall challenge can be solved, paving the way for further studies of applying raster-based deep learning models on map generalization.

POLICY HIGHLIGHTS

- Demonstrates the effectiveness of the proposed data structure with multiple evaluation indicators
- Identifies a “wobbly-wall effect” a challenge in deep-learning backed image based map generalization
- Proposes a layered data structure that separates a target building and its surrounding buildings to ease the learning task in training deep learning models for raster-based map generalization.

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Map generalization; deep learning; raster; building simplification; U-Net

Introduction

Map generalization is a cartographic process of deriving, from a source map, a target map at a reduced scale, while maintaining the source map’s main structural characteristics and information. There has been a long history of involving computers in the map generalization process with computer-assisted map generalization methods. Most of the existing map generalization methods, such as constraint-based methods (Ai & van Oosterom, 2001; Weibel & Dutton, 1998; Wei et al., 2021), mathematical morphology (Shen et al., 2019; Wang et al., 2022), and multi-agent based (Barrault et al., 2001; Ruas & Duchêne, 2007), are essentially rule-based, meaning that explicit cartographic rules or constraints are needed as part of the algorithm settings (Harrie & Weibel, 2007). These algorithms have partly been packaged as modules of

integrated platforms in commercial or open-source software (Barrault et al., 2001; Lecordix et al., 1997; Touya et al., 2019). These software packages are specialist systems that act as human-computer interactive (HCI) interfaces to assist cartographers. Cartographers have to use their expertise in cartographic rules to adjust the algorithms’ parameters to accomplish cartographic generalization for certain objects such as individual buildings, building groups, road segments, etc. Machine learning methods have been involved in deriving cartographic rules (Plazanet et al., 1998). Due to the limited capacity of conventional machine learning algorithms to generate image outputs directly, most machine learning algorithms are applied to categorize objects for different generalization operations, e.g. if an object should be retained, simplified, or eliminated (García Balboa &

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Ariza López, 2008; Karsznia & Weibel, 2018; Zhou & Li, 2016).

The recent success of deep learning (DL; Lecun et al., 2015) has attracted cartographic researchers to also apply deep learning neural networks to map generalization. The three major map generalization object classes addressed so far are buildings (Feng et al., 2019), roads (Courtial et al., 2020, 2022b), and rivers (Du et al., 2022). Researchers have experimented with DL architectures that are applicable for processing vector (Xiao et al., 2023; Yu & Chen, 2022; Zheng et al., 2021) and raster data. For building generalization, raster-based deep learning models such as Residual U-Net (ResU-Net) and Generative Adversarial Network (GAN) (Feng et al., 2019; Kang et al., 2020) have performed well in terms of global metrics such as positional accuracy. However, both pioneering studies reported an obvious issue that the generalized buildings have defects along their edges and in corners, exposing a *wobbly wall* effect, meaning that the rectangularity and parallelism of the buildings, and the straightness of building walls, cannot be preserved very well (Feng et al., 2019).

In this study, we apply raster-based deep learning architectures to a building generalization problem at large map scales that primarily involves the elimination of small objects, simplification, and aggregation operators. We propose a new layered data representation model that stores a target building for generalization in one channel and stores other surrounding buildings in another channel in the same tensor as the input for a neural network. We use the same deep learning network architecture used by Feng et al. (2019) as the backbone, composing a ResU-Net with the same raw map input, network hyperparameter setting, and evaluation metrics. This allows to focus our contribution on demonstrating the effectiveness of the proposed data representation model aligning with a new training set preparation strategy, thus addressing the wobbly-wall effect during the DL-based building generalization. We compare the new data model to the data model used in the previous studies, using additional evaluation metrics on geometry shape regularity.

We show that the straightness of the building boundaries can be well preserved visually and quantitatively with a combination of proper training sampling strategy and data representation model without changing the architecture of the commonly used deep learning neural network architectures or using a customized loss function. We also noticed that a larger patch size for the training samples is needed when the target map has a smaller resolution.

Related work

Conventional machine learning-based building generalization methods

Weibel et al. (1995) proposed that machine learning algorithms can assist in generating explicit or implicit knowledge for cartographic generalization in a human-in-the-loop interactive map generalization system. Machine learning algorithms such as discriminant analysis (Steiniger et al., 2008), decision tree (Steiniger et al., 2010), and artificial neural networks (Lee et al., 2017) are applied to learn rules for categorizing buildings in different geographical contexts to support the choice of map generalization operations such as selection/omission and simplification. It is noticeable that most applications of machine learning methods are limited to assisting human decisions by suggesting generalization operators on a specific building instead of outputting generalized buildings.

Deep learning-based building generalization methods

Vector-based models

Researchers have tried to generalize buildings as vector polygons from a larger map scale to a smaller map scale using DL neural networks in which graph convolutional neural networks (GCNN; Kipf & Welling, 2016) are the most widely used computational model family. Yan et al. (2019) applied GCNN to classify if a building group has a homogenous structure or not. Yan et al. (2020) used GCNN-backed autoencoder models to encode building shapes into letter descriptions automatically. Yan et al. (2022) further used GCNN to group buildings based on their spatial layouts. While the preceding works mainly focused on tasks of “structure recognition” (Brassel & Weibel, 1988) as a preparatory step of the core map generalization process, Zhou et al. (2022) were the first to propose an approach for vector-based map generalization, using a multi-task-based GCNN for simplifying the footprints of single buildings. All these advances provide evidence that deep learning models have the potential to also accomplish some critical steps in a vector-based map generalization workflow. However, an end-to-end solution has not yet been developed, as vector-based models cannot easily capture the spatial relationships between neighboring buildings yet. Mai et al. (2023) recently proposed a general-purpose representation model that enables a deep learning network to encode polygonal geometries as vectors, which sheds light on identifying spatial relationships of polygons. However, the model’s

capacity and efficiency in handling downstream tasks such as sensing the spatial layout of multiple geometric polygons have not yet been tested.

Raster-based models

Besides the vector-based models, more researchers have put effort into raster-based models, which conceptualize the map generalization process from a source map to a target map as an image-to-image translation process (Hertzmann et al., 2001; Long et al., 2015). Sester et al. (2018) is one of the first attempts to apply deep learning models for map generalization. The study applied a convolutional neural network (CNN)-backed model (namely U-Net; Ronneberger et al., 2015) to image patches for building generalization from 1:5,000 to 1:10,000, 1:15,000, and 1:25,000, and achieved very promising performance in terms of pixel-wise intersection-over-union (IoU). A follow-up study from the same group (Feng et al., 2019) applied additional models, including ResU-Net and GAN. Kang et al. (2020) also applied GAN for an end-to-end building generalization task from 1:10,000 to 1:25,000. Both studies show that DL neural networks have the potential to accomplish building generalization tasks but leave big room for improvement. One major deficit that both studies report is the wobbly wall effect, as the models can hardly generate straight walls in the output.

Different from the previous two studies that used binary images as the input and output for training, Courtial et al. (2022c) tried to integrate surrounding road networks of the buildings with a layered data representation model for building generalization between smaller scale maps (from 1:25,000 to 1:50,000). In this case, features such as buildings, roads, water bodies, and other semantic information, such as the alignment of buildings are stored in different raster channels of the input tensor for training deep learning neural networks. While the layered data representation model shows evidence of the benefit of providing extra cartographic constraints during the learning process, the output buildings still lack rectangularity and parallelism of building walls (Figure 1).

In summary, the raster-based deep learning models so far have shown a greater capacity to learn broader cartographic knowledge for building generalization than the vector-based models, as it is still a remaining challenge to develop topologically aware polygon encoders with deep learning architectures (Mai et al., 2021). However, it remains challenging to address the wobbly-wall effect in raster-based models. For this purpose, this work proposes a layered data representation model that separately stores a building for generalization and its

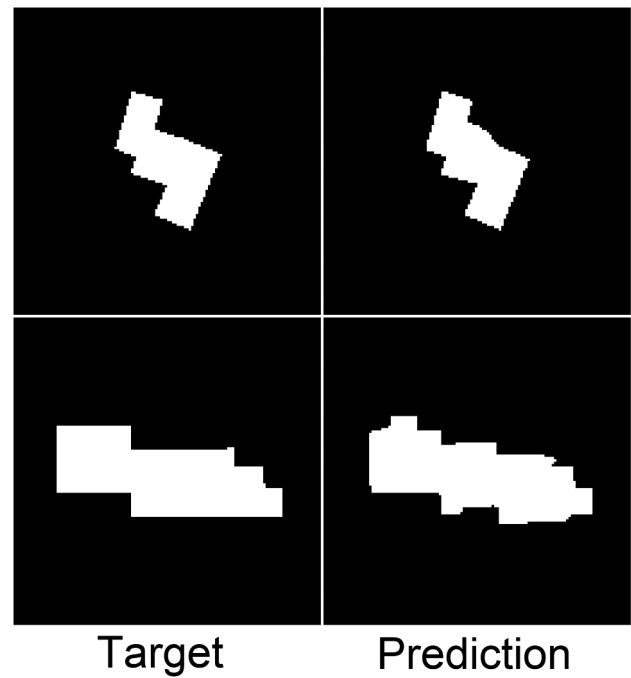


Figure 1. Samples of the “wobbly-wall effect” where predicted buildings have curvy walls with spurious protrusions or intrusions.

context buildings in different channels for learning to generalize raster maps.

Methodology

Overall workflow

As Figure 2 shows, we followed the same framework of Feng et al. (2019) that uses paired patches derived from the same spatial extent of both the source map and target map to train a deep learning neural network. Once the neural network is well trained and a new patch set (X) at the source map scale is input to the neural network, it is

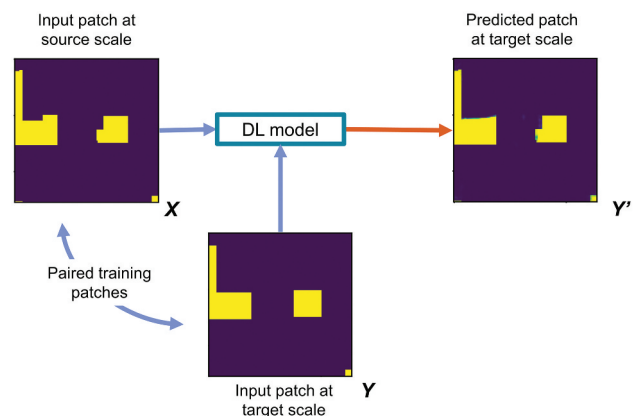


Figure 2. Conceptualized workflow.

Table 1. Factors that may influence the final performance of a DL task. Settings in bold are options whose impact on map generalization was investigated by this study.

Factor	Settings to refine in the factor
Prediction task	1:5,000 to 1:10,000, 1:5,000 to 1:15,000; 1:5,000 to 1:25,000; 1:10,000 to 1:50,000, etc.
Input patch	Data representation model Training set (patch size, number of paired patches, paired patch balance), etc.
DL model	Neural network architecture Loss function(s) Optimizer, learning rate, stop criteria, etc.
Predicted patch	Data representation model, training set, etc.

expected to generate an image set (Y') that represents the generalized map at the target scale. The model performance is evaluated by comparing the difference between the generated image set Y' and the real generalized map patch set Y .

A complete raster-based deep learning solution is mainly composed of four factors: prediction task, input patch, DL model, and predicted patch (Table 1). Each factor can have almost infinite options that can be improved to achieve a better result. Given limited time and computing resources, it is thus impossible to carry out a complete, systematic test covering all possible option combinations. In addition, different data sets may need individually refined neural networks. While the DL models have been investigated intensively in prior works, in this study, we mainly focus on improving the data representation of input patches and the strategies to derive paired patches from the source and target maps. We limit our investigation to map generalization transforms at two scale transitions only: from 1:5,000 to 1:10,000 and to 1:15,000, respectively, as highlighted in Table 1. We further explore the possible influence of several factors, such as neural network architectures.

Data

We used the same data set as Feng et al. (2019). The data set contains a series of binary maps at 1:5,000, 1:10,000, 1:15,000, and 1:25,000 of Stuttgart, Germany. The maps at scales smaller than 1:5,000 were generated from the 1:5,000 map utilizing the map generalization software CHANGE (Powitz, 1993), which is a traditional procedural software system implementing simplification and aggregation operators. The maps have already been split into a training map series within the same region ($42,800 \times 35,000$ pixels) and a test map series within another region of the same city ($4,270 \times 2,560$ pixels). No style or semantic information is applied to the maps. Therefore, the learning task for map generalization on

these maps is only based on geometric information of the existing buildings, without considering style, land use, or other geographical objects such as roads, forests, waterbodies, etc.

Building-centered layered data representation for learning

For the paired patches to train a deep learning neural network, most previous studies use single-channel, mutually exclusive map patches (Feng et al., 2019; Kang et al., 2020), creating segmentation boundaries such as the ones marked as red lines in Figure 3. Such a strategy may have two potential drawbacks: First, the mutually exclusive splitting strategy can only generate a relatively small number of paired patches, for example, around 32,000 patch pairs in Feng et al. (2019). Second, each patch pair contains several buildings, which makes the prediction task generalize a building set per patch simultaneously, rather than generalizing building by building. Conceptually, this may raise the difficulty of the prediction task as each spatial configuration of a few buildings can be almost unique.

Here, we propose a building-centered layered data representation for the map patches similar to Courtial et al. (2022c) and Wu and Biljecki (2022). We conceptualize a building generalization process as a per-building process: Each time, an algorithm has one *target* building to generalize, and the generalized output should consider not only the *target* building but also its surrounding buildings as the *context* that supports the decisions on which map generalization operators to apply to the target building (Figure 4). Therefore, for an individual building, we store the building and its surrounding building parts within a given extent in two channels of a tensor, respectively, and input the two-channel tensor to a deep learning neural network. Correspondingly, instead of generalizing all buildings within the spatial extent of the neural network's output, we only train the neural network to predict the generalized shape of the target building. Note that the layered data representation model could integrate further information, such as other geographic objects and geographical semantics into more channels of the input tensor in future work. Relying on this data representation model, we can extract sample patch pairs based on buildings, i.e. the yellow boundaries shown in Figure 3, which in most cases should generate more training patches (83,529 buildings in our dataset) than the mutually exclusive splitting strategy. In addition, because of the proposed data presentation model, per-patch evaluation is now the same as per-building evaluation. Such characteristic can help to refine training set construction strategies as well as training strategies, such as balancing training samples based on the

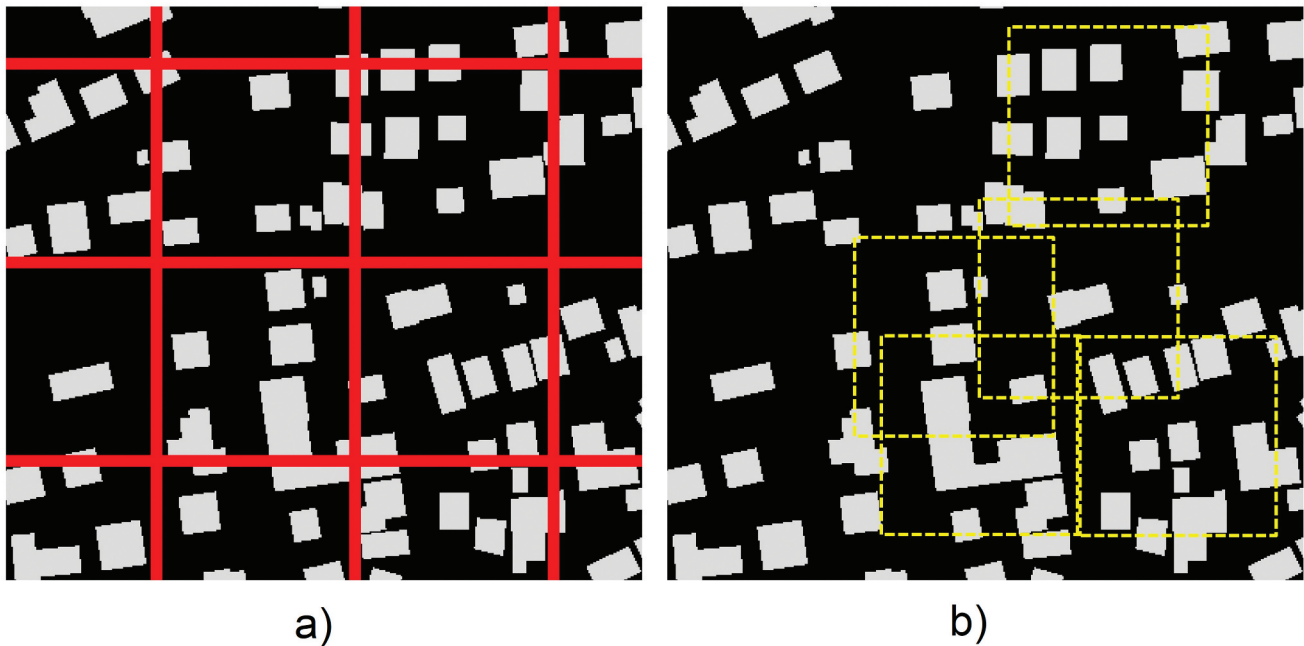


Figure 3. An illustration of a) The mutually exclusive map patches clipped by red solid lines and b) Samples of building-centered map patches, where each has a target building in the center, clipped by yellow dashed lines.

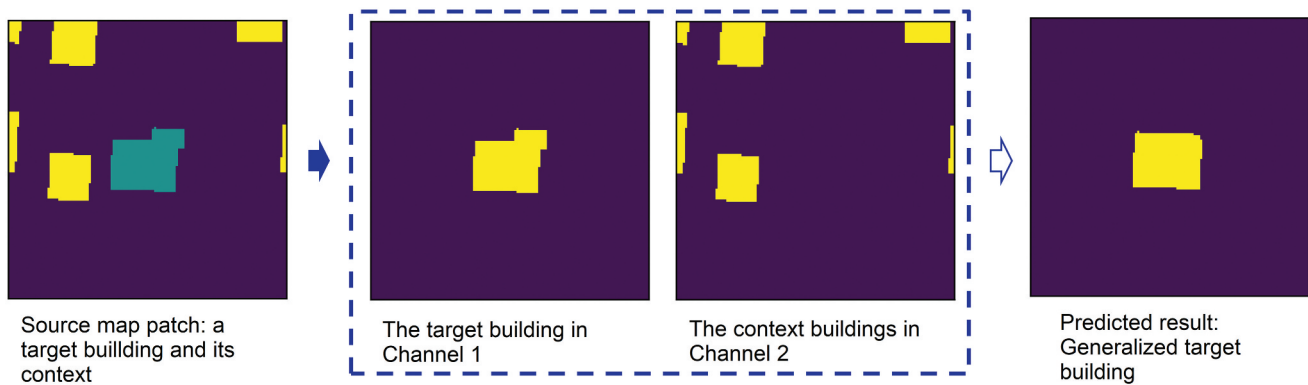


Figure 4. An illustration of the building-centered, layered data representation of a building and its corresponding prediction by a deep learning neural network.

difficulty of generalizing the buildings (e.g. making sure that a sufficient number of complex cases is represented in the training sample).

Residual U-Net and attention U-Net

There are controversial conclusions in the literature regarding which image-to-image translation deep learning neural network architecture is the best for map generalization. Feng et al. (2019) suggested that Residual U-Net (ResU-Net, Zhang et al., 2018) performs better than regular U-Net and GAN, while Courtial et al. (2022a) suggested that CycleGAN (Zhu et al., 2017) is better than U-Net and GAN-backed pix2pix model (Isola et al., 2017).

In this study, we choose two U-Net-backed neural networks: ResU-Net and Attention U-Net (AttU-Net, Oktay et al., 2018). A basic U-Net consists of an encoder and a decoder (Figure 5). The encoder has several convolution blocks applying convolution and down-sample pooling to an input tensor in a cascading way. During the cascading down-sampling, the width and height of the input tensor are halved while its depth is doubled. The decoder applies a reverse cascading process such that an input tensor is up-sampled by convolution filters to double its width and height while halving the depth. At the same level of the encoder and decoder, there is a skip connection that concatenates the output of an encoding block to the input of a decoding block. The improvement of ResU-Net compared to the original U-Net is to add a bypass within each

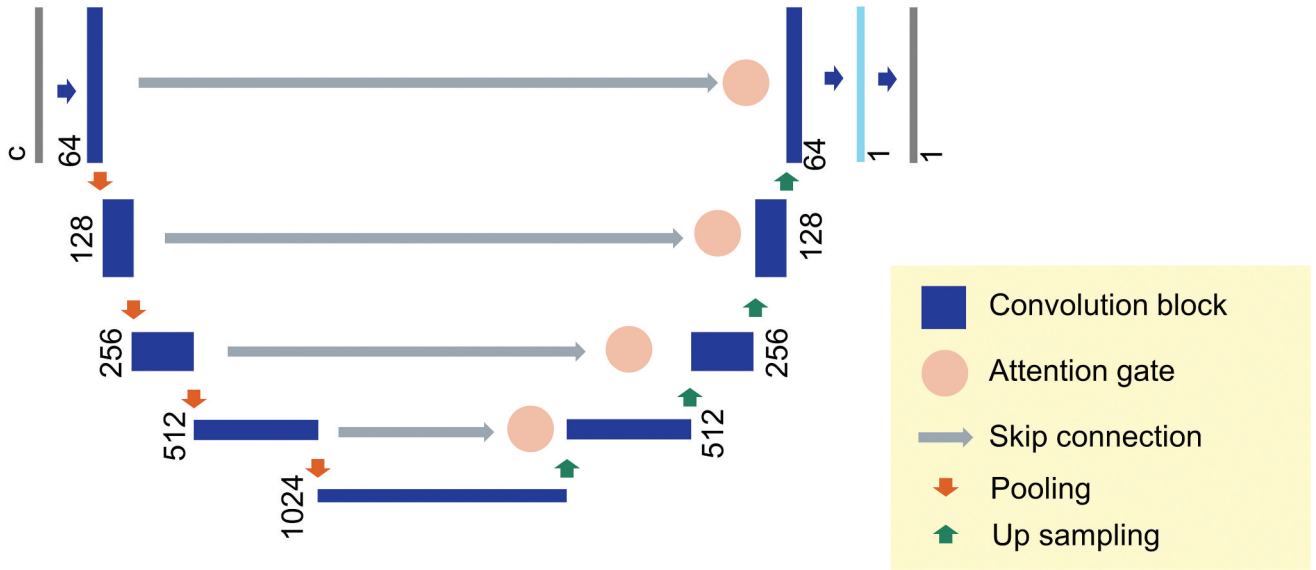


Figure 5. Illustration of an attention U-Net. Numbers denote the depth of the output tensor after convolution; c is the number of input channels that may vary between 1 and 2; the height and width may vary between 128×128 and 256×256 . Both are specified according to different settings. See details in the section on experiment settings.

convolution block that combines the input of the block with the initial output of the block, enabling the earlier layers in a deep network to be activated to address the issue of vanishing or exploding gradients, respectively, for better training (He et al., 2016). AttU-Net further improves U-Net by adding a self-attention gate to each between-block skip connection. The self-attention gates are expected to mimic cognitive attention and focus the neural network on specific important hidden patterns in the data (Vaswani et al., 2017). The initial output of the neural networks ranges between 0 and 1. We binarized the final outputs by setting the threshold at 0.5.

We primarily used mean squared error (MSE, Equation 1) as the loss function for training the neural networks:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (1)$$

where y_i is the true value of a pixel and $\hat{y}_i$ is the predicted value of the same pixel. MSE is suggested as the best loss function in the empirical experiments in Feng et al. (2019).

In addition, we tried focal loss (Lin et al., 2017) for some neural networks as well:

$$FocalLoss(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t) \quad (2)$$

where p_t is a measurement of the prediction accuracy for the pixel with label $y = 1$, α_t and γ are scaling parameters to moderate imbalance. Focal loss is designed for imbalanced samples, which fits many of

our training cases as the building pixels are usually much fewer than the background pixels in a patch.

Evaluation metrics and strategies

Several common metrics used in previous studies were chosen to evaluate the model performance, including *IoU*, *overall accuracy*, *precision*, and *recall*. Among the metrics, we paid more attention to *IoU* as it is more sensitive in map generalization tasks (Sester et al., 2018). We further used *rectangularity*, *fractality*, and *roughness index* (Basaraner & Cetinkaya, 2017) as proxies for quantitatively measuring the regularity of the building geometry shapes. Rectangularity is the areal ratio between a building geometry shape and its minimum area bounding rectangle (MABR). As demonstrated in Figure 6a,b, A rectangle-like shape can fit its MABR very well, with an *IoU* close to 1. Thus, rectangularity can be an indicator of regularity. Mathematically, rectangularity equals the *IoU* of the building shape and its MABR, as their intersection is always the building itself, and their union is always the MABR. In addition, comparing the MABRs of the target and predicted buildings (Figure 6c,d), it is suggested that the *IoU* of MABR boundaries can be more sensitive to the difference. We thus used MABR Boundary *IoU* and MABR *IoU* as the primary proxies for evaluating regularity.

Fractality and roughness measure edge roughness as the secondary proxy for the regularity (Equations 3 and 4), where A_{PN} is the area of a building polygon, P_{PN} is the perimeter of the polygon, and $\mu_{r_{ibp}}$ is the mean distance from each boundary pixel to the centroid of

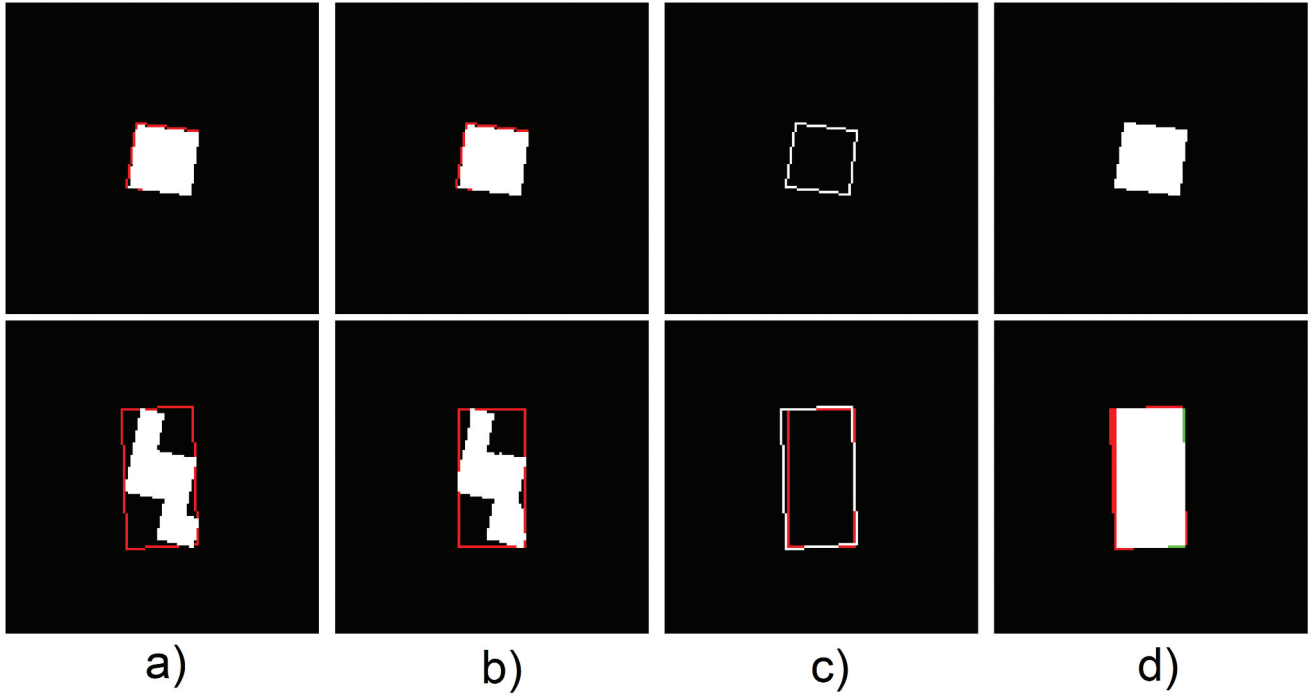


Figure 6. Illustration of geometry shapes in map generalization, their minimal bounding boxes, and IoUs. a) a building in a target map (1:10,000) and its minimal bounding box (in red); b) A building in a predicted map (1:10,000) and its minimal bounding box (in red); c) Boundaries of the target minimal bounding box (in white) and the predicted minimal bounding box (in red); d) Intersection (in white) and union (in nonwhite color) of two minimal bounding boxes (target building in the red channel, predicted building in the green channel, and the intersection of the target and predicted buildings in the blue channel).

the polygon. To measure the similarity between the predicted building and the target building on these two metrics, we used the absolute difference. Thus, a smaller average absolute difference indicates a better result.

$$FR = 1 - \frac{\log(A_{PN})}{2 \times \log(P_{PN})} \quad (3)$$

$$RI = \frac{\mu_{r_{ibp}}^2}{A_{PN} + P_{PN}^2} \times 42.62 \quad (4)$$

One of our map-splitting strategies to create training patch pairs is the building-centered strategy. Our map patches potentially overlap with each other, which is different from previous studies. At the same time, all cropped map patches have their lower corner in terms of row number and column number in the map recorded as their file name. The prediction of test patches thus can be merged back to a whole prediction map again. Therefore, for a comprehensive understanding of the model performance, we constructed a predicted map by merging the predicted patches with a conflict-solving rule that if any predicted value of overlapping pixels is one,

the pixel value in the predicted map is set as one. Then, the metrics are applied to compare the predicted map to the real generalized map. In addition, because we excluded buildings larger than the patch size when preparing patches with the building-centered models, the building sets involved in different data representation models are slightly different. Therefore, we also applied the metrics separately to each generalized building predicted by the trained neural network.

Experiment settings

Each experiment is a combination of data representation models, patch sizes, prediction tasks, and deep learning neural networks. Each deep learning neural network further involves model-specific hyperparameters for fine-tuning, including learning rate, batch size, type of attention head, etc. Table 2 shows that the data representation and input-output models already lead to four options. Data Model (DM) 1 represents the baseline model of Feng et al. (2019). DM 4 is our main proposal that combines the building-centered patch splitting strategy and multi-channel data representation. DM 2 and DM 3 are intermediate solutions

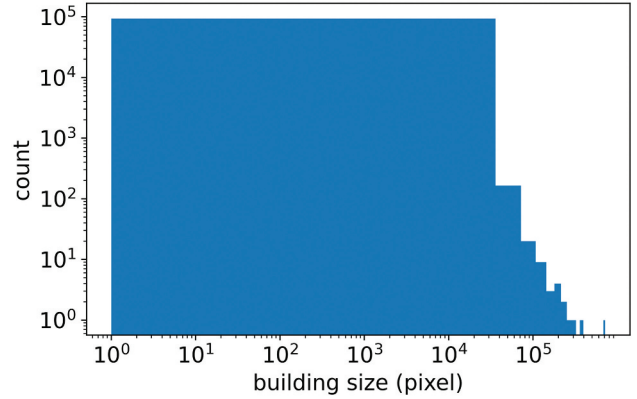
Table 2. Summary of the data representation models used in the experiments.

DM ID	Data model	Input	Prediction	Patch splitting
1	Single channel	Buildings	Buildings	Mutually exclusive, as Feng et al. (2019)
2	Single channel	Buildings	Buildings	Building centered
3	Single channel	Buildings	A building	Building centered
4	Multi-channel	Ch 0: a building Ch 1: context	A building	Building centered

between DM 1 and DM 4 that can explore if the building-centered patch splitting strategy or the multi-channel data representation model contributes more to a better generalization result.

Because of the splitting strategies, the training and test patch sizes for different data models are different (Table 3). We chose the scale transitions from 1:5,000 to 1:10,000 and 1:15,000, respectively. The primary patch size is 128×128 pixels. To investigate the influence of patch size on model performance, we also prepared one building-centered patch set (DM 4) with 256×256 pixels. For building-centered patches, if the building is bigger than the patch size, the cases were excluded. Therefore, a larger patch size (256 pixels) includes slightly more valid buildings.

From an initial scan of the building size in terms of the number of pixels, we observed a high imbalance between small buildings and large buildings (Figure 7). Thus, there is a potential concern over imbalanced training as generalizing small buildings is relatively easier between the scales we investigated. Increasing training cases with complex generalization processes may help increase a neural network's capacity for generalization. We thus employed the pixel difference between the source building and the target building as a simple proxy for the complexity of a generalization situation, with the assumption that a larger difference in terms of pixels may involve a more complicated generalization process, e.g. a combination of several generalization operators or a more complex geometric transformation of the shape. We further augmented those patches with greater changes into five augmented patches by flip (vertical and horizontal flips) and rotation (90-, 180-, and 270-degree rotations) operators for the building-centered 128×128 patch set if the center building of a patch is larger than 200 pixels.

**Figure 7.** Building size distribution of the source map at 1:5,000.

The neural networks were adopted from an online repository¹ implemented in PyTorch (Paszke et al., 2019) and trained with a single NVIDIA V100 Tensor Core GPU.

Results

Results for the generalization from 1:5,000 to 1:10,000

During training, all models showed a constant difference between their training loss and validation loss (Figure 8). This difference might be partially due to the complex generalization scenarios that make the training set and validation set slightly different. A generalization scenario may involve one individual generalization operator or a combination of several operators, which leads to many potential scenarios. Because building distributions in the training set and validation set are different in terms of building shapes, sizes, and other geometric properties, sample distributions of generalization scenarios in the training set and

Table 3. Patch size and corresponding training and test sets regarding different map splitting strategies.

DM	Patch size (pixel)	Splitting	1:5k $\rightarrow$ 1:10k			1:5k $\rightarrow$ 1:15k		
			Training	Valid.	Test	Training	Valid.	Test
1	128×128	Mutually exclusive	32,255	3,583	431	32,059	3,562	430
2,3,4	128×128	Building centered	83,529	9,281	1,548	83,529	9,281	1,548
4	256×256	Building centered	-	-	-	85,817	9,535	1,544
4	128×128 augm.	Building centered	-	-	-	83,529 + 11,402	9,281	1,548

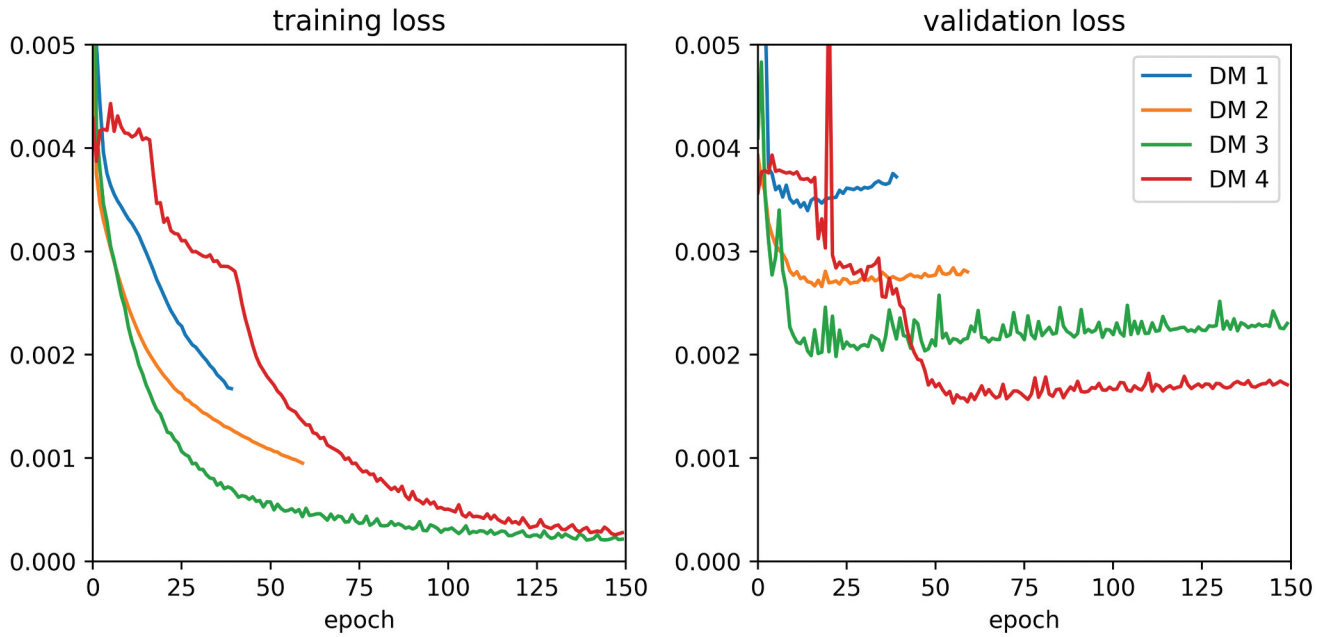


Figure 8. Training loss and validation loss of ResU-Net models with different data models.

validation set also cannot perfectly match. We chose models whose difference between the validation loss and the training loss remains stable. Neural networks for DM 1 and DM 2 had an earlier stop in terms of epochs because the difference between their training loss and validation loss showed an enlarging trend of overfitting. In general, we can observe training loss remained low with all data models, trending to a similar minimum bound.

For the final output as a whole map, our proposed data representation model (DM 4) performs the best in three general metrics at both the whole-map level (Table 4) and

per-building level (Table 5). For the performance on individual buildings, DM 4 also has better results than the other DMs. Compared to Feng et al. (2019), the most noticeable improvement of our proposed model is on IoU, with p -value <0.001 , <0.001 , <0.05 against DM 1, DM 2, and DM 3 in the Mann-Whitney U-test, respectively. For predicting multiple buildings, increasing the training size (DM 2) can achieve some improvement in the overall accuracy and IoU, compared to the results of DM 1. Similar results apply to MABR IoU, MABR Boundary IoU, absolute differences of fractality and roughness as well (Appendix 1).

Table 4. Per-pixel metrics of final map reconstructed from predicted patches by ResU-Net on 1:5,000 to 1:10,000 generalization with 128×128 pixel patches. Cells are colored by their z-score regarding the metric. Dark blue: <-1.5 ; light blue: $-1.5 \sim -0.5$; light yellow: $-0.5 \sim 0.5$; light red: $0.5 \sim 1.5$; dark red: >1.5 .

Metrics	DM 1	DM 2	DM 3	DM 4
IoU	0.9671	0.9745	0.9608	0.9807
Accuracy	0.9952	0.9966	0.9952	0.9977
Precision	0.9852	0.9810	0.9881	0.9894
Recall	0.9814	0.9933	0.9720	0.9911

Table 5. Mean and standard deviation of per-building metrics of predicted results by ResU-Net on 1:5,000 to 1:10,000 generalization with 128×128 pixel patches. Numbers in bold are the best results of each metric.

Metrics	DM 1	DM 2	DM 3	DM 4
IoU	0.9720 (0.0919)	0.9702 (0.0974)	0.9702 (0.1020)	0.9809 (0.0756)
Accuracy	0.9974 (0.0161)	0.9976 (0.0088)	0.9974 (0.0112)	0.9984 (0.0065)
Precision	0.9875 (0.0560)	0.9737 (0.0929)	0.9859 (0.0631)	0.9876 (0.0638)
Recall	0.9840 (0.0693)	0.9964 (0.0117)	0.9842 (0.0759)	0.9936 (0.0298)
MABR IoU	0.9659 (0.1047)	0.9550 (0.1161)	0.9627 (0.1150)	0.9739 (0.0915)
MABR Boundary IoU	0.8056 (0.2992)	0.7548 (0.3185)	0.8082 (0.3086)	0.8354 (0.2803)
Abs. Δ FR	0.0019 (0.0072)	0.0016 (0.0040)	0.0018 (0.0072)	0.0011 (0.0035)
Abs. Δ RI	0.0146 (0.0432)	0.0161 (0.0466)	0.0128 (0.0387)	0.0096 (0.0358)

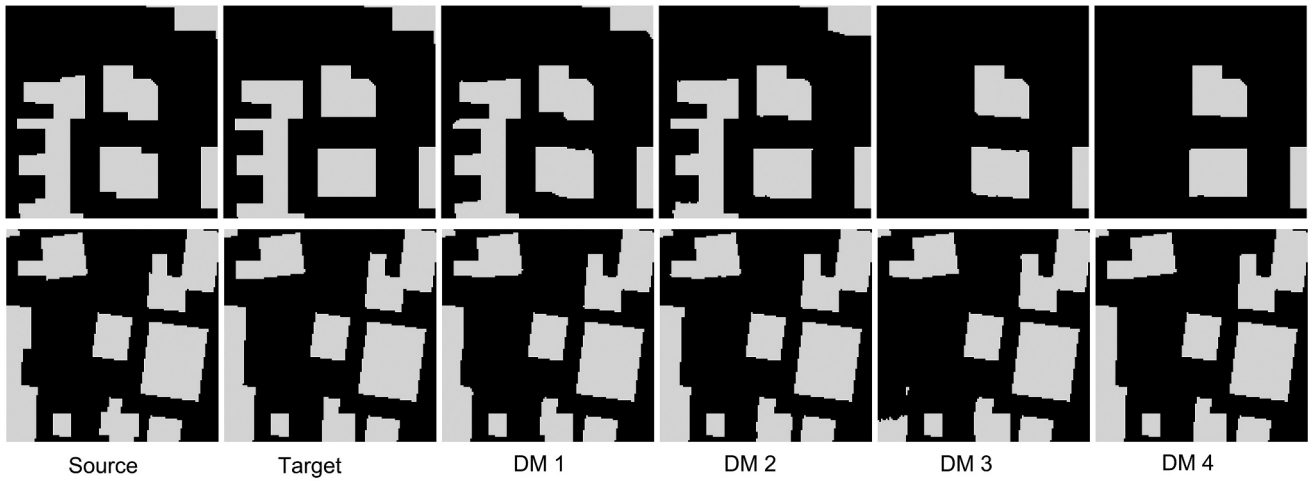


Figure 9. Two sample areas cropped from the source, target, and the reconstructed predicted maps. The image size is 128×128 pixels. The images are not initial patches for training and testing.

From randomly selected patch samples from the reconstructed generalized maps (Figure 9), it can be observed that all models predict shapes that are roughly similar to the target. For the data model (DM 1) from Feng et al. (2019), the wobbly wall effect remains for walls, corners, and the endpoints of some long protrusions, such as in Buildings A and B. For the output buildings in single-building prediction models (DM 3 and DM 4), the walls are quite straight without many protrusions along the boundary. The corners have much fewer fault pixels that make the corners look more mitered than rounded. Combining the two characteristics, the geometric shapes of the buildings are more realistic than in previous studies (Feng et al., 2019; Kang et al., 2020). For example, DM 4 has much fewer fault pixels, such as in Buildings A, B, and C. In some cases, we observe model break-downs, such as on Building D for DM 3, where a major building part is missing. Another noticeable difference is that for the final maps of DM 3 and DM 4, large buildings are missing, such as Building E. That is because the large buildings

were omitted for DM 3 and DM 4. More examples are available in Appendix 2.

Results for the generalization from 1:5,000 to 1:15,000

Regarding the generalization from 1:5,000 to 1:15,000 (Table 6), all the data models have a drop in the evaluation metrics compared to the task of generalizing from 1:5,000 to 10,000, with the same hyperparameter settings of the neural networks used for the generalization task from 1:5,000 to 1:10,000. In addition, DM 4 is not the best choice among the four data models.

All five methods that can potentially help to improve model performances do have some contribution to DM 4 for certain metrics. A ResU-Net with attention (DM 4 AttU-Net 128) is a more complex neural network than the original ResU-Net and should have the capacity to model more complex patterns. The complex model has performance increases on IoU, accuracy, and recall but a tradeoff in precision. Data augmentation on complex

Table 6. Per-pixel metrics of the final map reconstructed from predicted patches by variants of ResU-Net on 1:5,000 to 1:15,000 generalization. Cells are colored by their z-score regarding the metric. Dark blue: < -1.5 ; light blue: $-1.5 \sim -0.5$; light yellow: $-0.5 \sim 0.5$; light red: $0.5 \sim 1.5$; dark red: > 1.5 .

	DM 1	DM 2	DM 3		DM 4					
					ResU-Net		ResU-Net			
Metrics	ResU-Net				AttU-Net		augm	focal	ResU-Net	AttU-Net
Patch size	128					256				
IoU	0.9447	0.9549	0.9257	0.9136	0.9213	0.9452	0.9446	0.9659	0.9434	
Accuracy	0.9918	0.9938	0.9908	0.9893	0.9900	0.9932	0.9932	0.9954	0.9922	
Precision	0.9758	0.9642	0.9758	0.9694	0.9429	0.9674	0.9811	0.9802	0.9549	
ecall	0.9674	0.9900	0.9474	0.9407	0.9758	0.9763	0.9620	0.9851	0.9874	

patches (DM 4 ResU-Net 128-augm) follows a similar pattern as the DM 4 AttU-Net 128 with improved IoU, overall accuracy, and recall. However, it has a tradeoff in precision as well. When the neural network replaces MSE with focal loss (DM 4 ResU-Net 128-focal), all metrics achieve an increase. Enlarging the patch size to 256×256 pixels (DM 4 ResU-Net 256) also increases performance in all four evaluation metrics and has the best performance among the three methods. However, combining two performance techniques (attention head and enlarging the patch size in DM 4 AttU-Net 256) does not achieve the best performance in reconstructing the test map. Nevertheless, the five methods that theoretically can improve a neural network's performance are effective at least on certain metrics. As a reference, the results of the four methods, in terms of IoU, are also better than the previous study testing on the same task for the same data set (IoU = 0.9395 reported in Sester et al., 2018).

From the randomly selected patch samples (Figure 10), models with better general metrics (DM 4 ResU-Net 256 and DM 2 ResU-Net 128) produce visually better results than other models in general. Models can certainly process simplification for simple-shaped buildings, such as most rectangle- and square-shaped buildings in the samples. However, complex buildings such as Building A, C, and E are not well generalized by all models. Neural network with DM 2 fails in the simplification of Building A and Building E. DM 4 with focal loss (DM 4 ResU-Net 128-focal) fails in the simplification of Building C. Several models (DM 3 ResU-Net 128, DM 4 ResU-Net 128, DM 4 ResU-Net 128-focal, and DM 4 ResU-Net 256) process the non-parallel walls of Building A quite well, suggesting the models do not just learn simple rules such as rectangularization.

For two-building aggregation tasks such as on Building G, all models perform well. However, for multi-building aggregation such as on Building D, all models fail to distinguish the small detached building from the large building next to it. For the aggregation to make the large building, it can also be observed that the walls of model outputs still have varying degrees of defects.

Discussion

Data representation models and learning tasks

Given the sampled prediction patches of two generalization tasks (from 1:5,000 to 1:10,000 and from 1:5,000 to 1:15,000), we can observe that straight walls as predicted results are certainly possible, which is against the observations in previous studies. As the buildings are simple geometries, CNN-backed

architectures such as U-Net should have the capacity to learn the patterns as previous CNN models have shown (Shang et al., 2016). That is, simple geometry components such as straight lines and arches are learned in lower neural layers, and geometric shapes such as rectangles are represented in higher neural layers as combinations of the simple geometry components. The higher evaluation metric values of Data Model 2 compared to the metric values of DM 1 in Tables 4 and 5 suggest that insufficient training samples are one reason for the wobbly wall effect.

We also concluded several other reasons that can contribute to better predictions on map generalization. The layered data representation model, which separates a centered building from its surrounding neighbors and only applies neural network prediction to the centered target building, which we proposed as Data Model 4, further simplifies the prediction task and increases the possible training sample size. Based on the comparison of results, we see it as another reason contributing to reducing the wobbly-wall effect, as the prediction loss will not be averaged among other buildings during the training process.

Using larger sizes for input and output patches helped to improve model performance for the same architecture, as Table 6 shows. However, it also came at the cost that the model size increases, consuming more graphics memory, such that only reducing batch size for training can avoid running out of graphics memory, which further leads to much longer training times (about 12 min per epoch for 128×128 input vs. about 17 min per epoch for 256×256 input for the ResU-Net architecture).

The attention mechanism in deep learning was believed to improve the model performance significantly but showed limited effect in the experiments with the layered data representation model (DM 4). A possible explanation is that our centered target building is a manual attention mechanism forcing the neural networks to pay attention to the central area of the patches. Adding an attention mechanism to the regular ResU-Net also has an obvious tradeoff that the training time for each epoch also increases significantly. Although the attention mechanism does not show a significant improvement for the current experiments, it is still a candidate for future tasks of map generalization at smaller map scales where more complex aggregation, typification, and exaggeration will be involved.

Besides the detailed training process regarding data representation models and neural network structures, the complexity of generalizing the test map is also a concern. We observe that bad prediction performance, in terms of a lower IoU between the prediction and

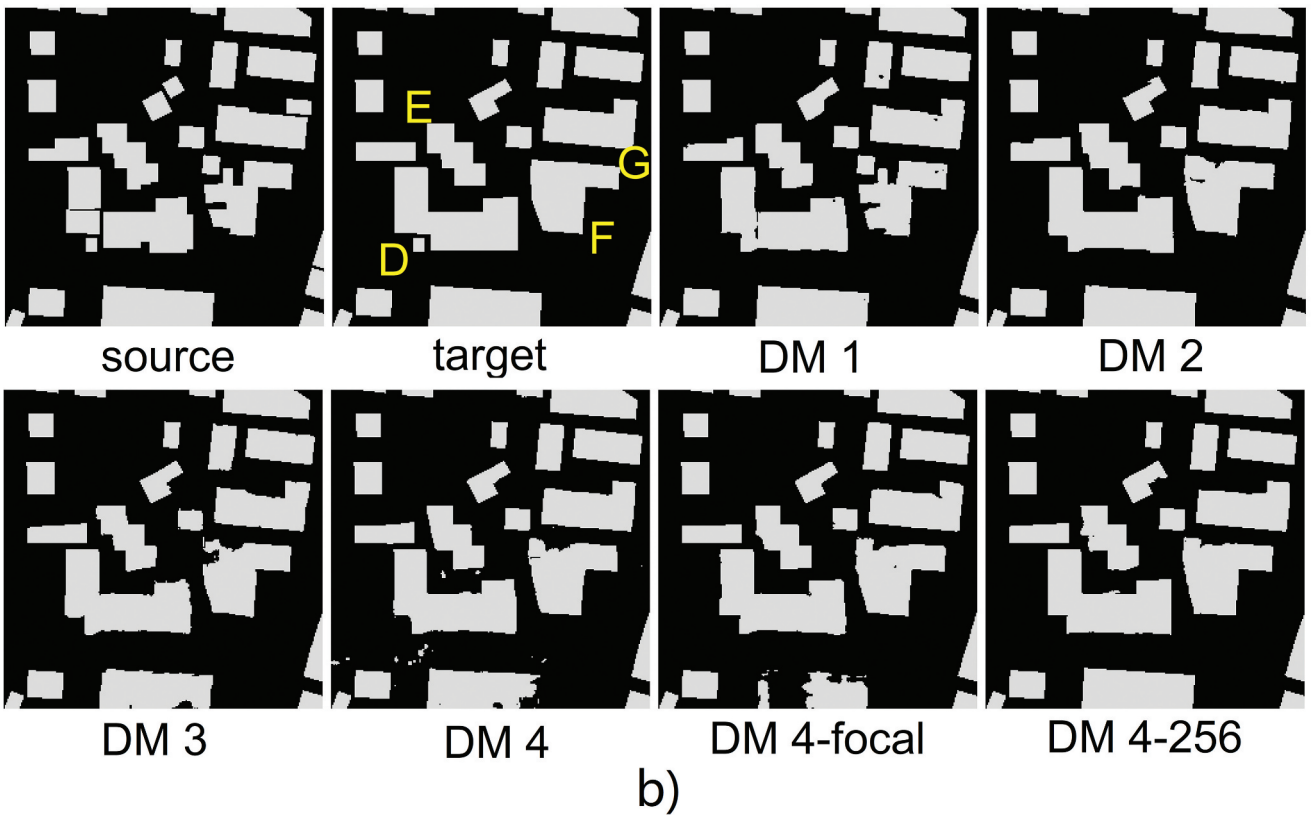
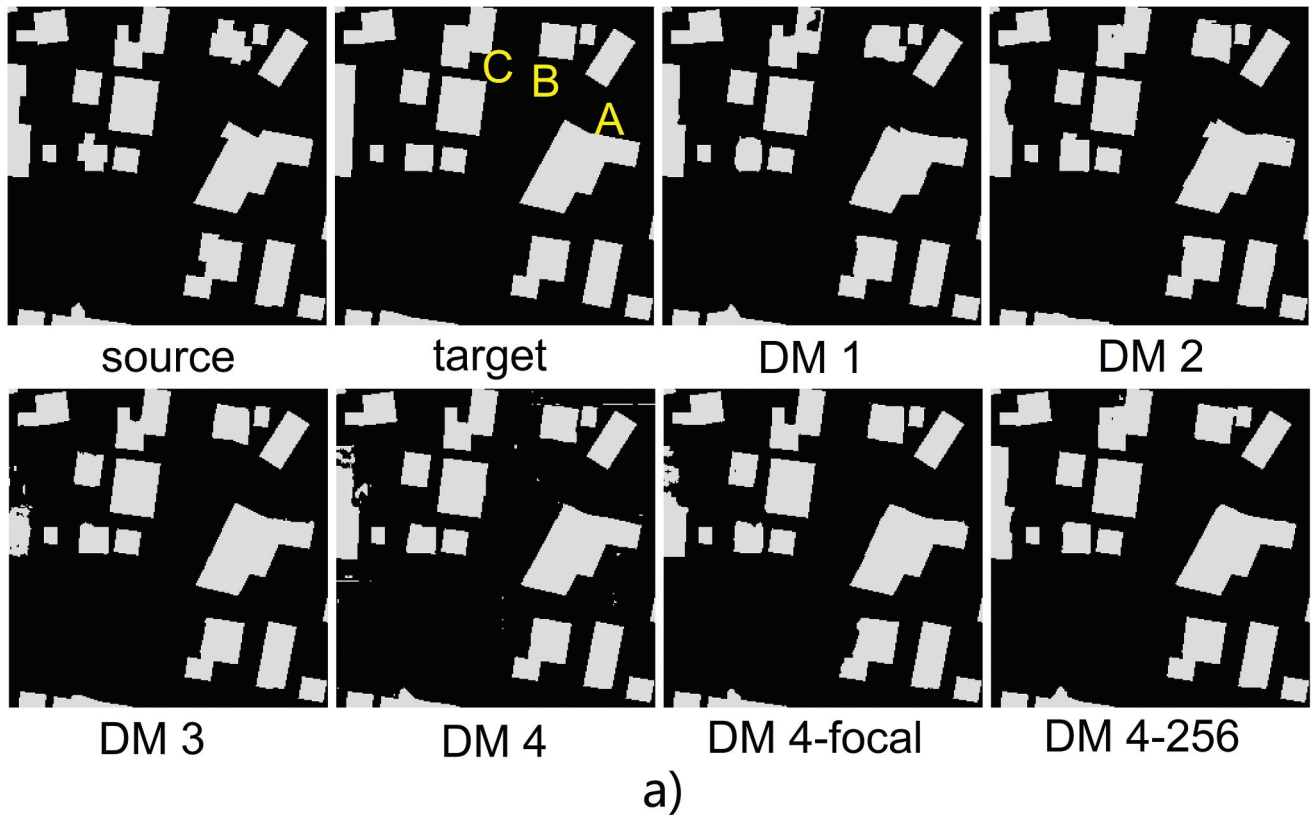


Figure 10. Source, target, and predicted maps for two sample areas. The image size is 256×256 pixels. Annotations correspond to the items in Table 6.

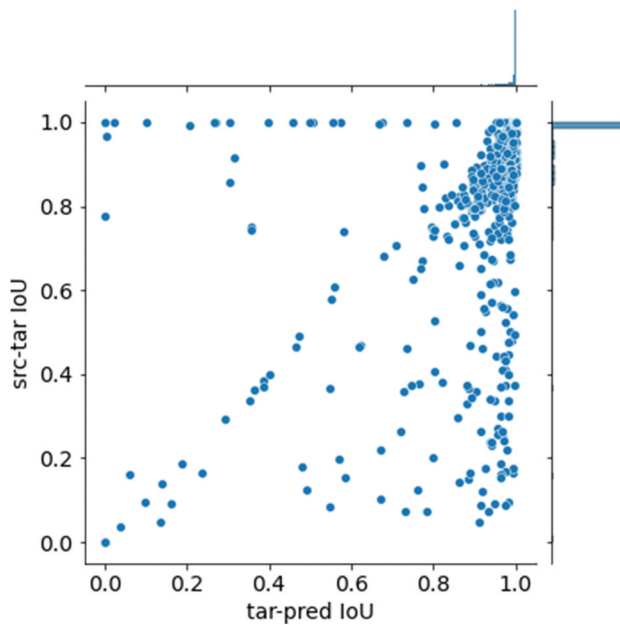


Figure 11. The IoU of source-target against the IoU of target-prediction on test patches of DM 4 ResUNet 256 for the building generalization from 1:5,000 to 1:15,000.

target patches, mainly corresponds to a lower IoU between the source and target patches (Figure 11), meaning these patches have a complex pattern to learn. It is also clear that the majority of the patches have an easy transformation pattern to predict, as the per-patch source-target IoUs mainly fall into the range between 0.8 and 1.0. This pattern also fits the reality that most buildings in Stuttgart are single- or multi-family houses with a simple geometric shape. Combining the observation that the simple data augmentation process on patches with possible complex patterns helped to improve the ResU-Net (Table 6), we propose to include more complex strategies in the future, such as bagging different neural networks that are trained on subsets that are categorized by the degree of changes or generalization operators between source and target patches, augmented with customized strategies.

Limitations

This paper focused on exploring solutions for solving the aforementioned wobbly-wall effect reported in previous studies. Therefore, the map generalization tasks in the reported experiments are limited to large map scales (1:5,000 to 1:10,000 and 1:5,000 to 1:15,000), which primarily involve simplification and aggregation operators only. Our current solution thus only works for simplification and easy aggregation tasks without checking its performances on map generalization for smaller scales in which more complicated generalization

operator combinations such as typification may be conducted and wider and more complex geographical contexts may be involved. In addition, this study also does not take into account other geographic objects and semantics, such as roads and land use information, for map generalization.

Besides the limitations of the map generalization tasks, we also could not fully explore the model performances of all possible combinations of input-output models and neural network architectures. Because of the limited tensor size that a GPU can process, we excluded buildings with a size larger than a patch can represent, as splitting a large building into several patches loses the overall geometric information as well as the whole context around the building. Therefore, this strategy excludes large buildings, given the current patch sizes. Further fine-tuning on neural network parameters, such as another loss function, may also provide a better result than our current solution. In addition, we chose the epoch when the gap between the training loss and validation loss was stable without further significant enlarging for all models, following a general principle for identifying the overfitting issue (Ying, 2019). However, the distributions of the training set and the validation set can be slightly different regarding the map generalization operators applied to the buildings in each set. That is due to the complex nature of map generalization: Each individual map generalization operator, such as simplification, aggregation, typification, and displacement, and their binary and ternary combinations can be treated as a unique subtask to learn by the neural network. However, it is very unlikely that the samples for each subtask in the randomly generated training set, validation set, and test set follow the same distribution. Therefore, the gap between the validation loss and training loss may not be the best metric for indicating overfitting in the scenario of map generalization. We need further investigation to create a balanced training set and a robust validation set for improved training of the neural networks. However, this is a non-trivial issue due to the aforementioned potential combinations of map generalization operators.

The output comparisons between the proposed model and the benchmark models are based on the best instances within their own individual training process. The reliability of each instance regarding the randomness of hyperparameters is not fully explored.

Conclusions

In this study, we have focused on improving raster-based deep learning models for building generalization at large map scales, where the key generalization

operators involved include elimination of small objects, simplification, and aggregation. Our work has led to the following main contributions:

- We proposed an adapted, layered data representation model for the input and output building patches in deep learning-backed map generalization workflows.
- As our building generalization experiments with different settings on a data set used in a previous study have shown, the new layered single-building centered data representation model can help increase the neural networks' performance on map generalization tasks.
- Particularly, the wobbly-wall effect commonly observed in previous studies is significantly reduced. The predicted buildings are visually more realistic, maintaining rectangularity, parallelism, and straight walls. This further suggests that raster-based deep learning architectures still have a great potential in the application field of map generalization.
- Comparisons of the experimental results further suggest that increasing the patch size for the proposed data representation model can increase model performances better than adding more complex neural network elements such as attention heads.

A few new challenges were observed during the study that should be solved in future work. We will further explore the distribution of different map generalization operators and their combinations in different map scales and build a more balanced training set. The model's reliability regarding the randomness in the value initialization of hyperparameters needs to be examined. We also plan to check the effectiveness of the proposed solution on official maps that are validated by professional cartographers.

Note

1. https://github.com/LeeJunHyun/Image_Segmentation

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Data availability statement

The raw maps that support the findings are available by request to Dr Yu Feng (y.feng@tum.de). The codes for U-Net and its variants are from third-party authors who are not affiliated with this manuscript. The codes for data preprocessing and the models adapted from U-Net models are available here: <https://doi.org/10.6084/m9.figshare.21901086.v1>.

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